UNIVERSITY OF SINDH JAMSHORO

DEPARTMENT OF COMPUTER SCIENCE

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[2K23/CSM/132]

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Course Name: [ARTIFICIAL INTELLIGENCE]
Date: [6/24/2026]

Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence
Topic: AI Tutor for Personalized Student Learning Recommendations

Task 1: PEAS Specification

Performance Measures

- Student learning improvement
- Recommendation accuracy
- Student engagement rate
- Response time

Environment

Online learning platforms, classrooms, mobile applications, internet connectivity, students with different learning styles and educational content.

Actuators

Personalized lesson recommendations, quizzes, feedback messages, notifications, study plans.

Sensors

Student quiz scores, assignment results, learning history, clickstream data, time spent on lessons.

Task 2: Environment Classification

Observability: Partially Observable
Justification: The system cannot directly observe a student's true understanding.

Agents: Single-agent
Justification: The AI tutor is the primary decision-making agent.

Determinism: Stochastic
Justification: Student behavior and learning outcomes are uncertain.

Episodic/Sequential: Sequential
Justification: Current recommendations affect future learning.

Static/Dynamic: Dynamic
Justification: Student performance changes continuously.

Discrete/Continuous: Discrete
Justification: Recommendations and learning activities occur as separate events.

Task 3: Critical Analysis

1. The greatest challenge is the stochastic nature of the environment because students learn differently and their performance is unpredictable. The AI must continuously adapt its recommendations using new learning data.

2. Utility Function: U = 0.7 × Learning Accuracy + 0.3 × Student Engagement. If the weight of learning accuracy is doubled, the AI will prioritize more accurate recommendations even if they reduce engagement slightly.

Task 4: Structured Representation (JSON)


```
{
 "system_name":"AI Tutor",
 "peas":{
"performance_measures":["learning_improvement","recommendation_accuracy","engagement_rate","response_time"],
   "environment":"Online learning platform with students and digital content",
   "actuators":["lesson_recommendations","quizzes","feedback","notifications"],
   "sensors":["quiz_scores","assignment_results","learning_history","time_spent"]
 },
 "environment_classification":{
   "observability":{"choice":"Partially Observable","justification":"True knowledge cannot be observed directly."},
   "agents":{"choice":"Single-agent","justification":"The AI tutor makes the decisions."},
   "determinism":{"choice":"Stochastic","justification":"Student behavior is unpredictable."},
   "episodic":{"choice":"Sequential","justification":"Current actions influence future learning."},
   "dynamic":{"choice":"Dynamic","justification":"Student progress changes over time."},
   "discrete":{"choice":"Discrete","justification":"Recommendations are delivered in separate steps."}
 },
 "utility_function":"U = 0.7*learning_accuracy + 0.engagement"
}
```